ARTIFICIAL INTELLIGENCE

Semester	III				
Course code					
Category	Professional Core Courses				
Course title	Artificial Intelligence				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVE:

1. Understand the very basics and Uses of Artificial Intelligence (AI)
2. Understand the concept of Intelligent agent and its environment.
3. To provide the most fundamental knowledge to the students so that they become familiar with basic principles of AI towards problem solving, inference, knowledge representation and learning
4. Understand the logic-building methods and inferences for the knowledge representation.
5. Explore application of AI techniques in Expert systems, Neural Networks.

UNIT - I

Introduction to AI: What is AI, Turing test, History of AI, Artificial Intelligence Techniques, advantages, and limitations of AI, Impact and Examples of AI

Applications of AI by domain: Transportation, home/service robots, healthcare, education, low-resource communities, public safety and security, employment and workplace, entertainment, finance, banking and insurance.

Intelligent agent: Nature of Agents, Rationality and Rational agent with performance measures Flexibility and Intelligent agents, Task environment and its properties, Types of agents, other aspects of agents.

UNIT – II

Problem solving methods: Problem Solving Approach to Typical AI problems

Searching techniques: Uniformed search- General search Algorithm, Uniformed search Methods-Breadth first Search, Depth first search, **Informed/Heuristic search**- Generate and test, Best First search, A* Algorithm, AO* research, **Local search Algorithms**-Hill Climbing, Simulated Annealing, Genetic Algorithms, **Game as a search problem**-Minimax approach, Minimax Algorithm, Alpha beta pruning, Constraint satisfaction problems (CSP).

UNIT - III

Knowledge Representation schemes and reasoning: Approaches and issues, procedural vs declarative knowledge, Matching, conflict resolution.

Logic: Propositional logic, predicate logic, Resolution, Resolution in proportional logic and predicate logic, Clause form, unification algorithm.

Uncertain Knowledge and reasoning: Methods, Bayesian probability and belief network, Probabilistic reasoning, Forward and backward reasoning, Dempster -Shafer theory.

UNIT - IV

Planning: The Planning problem, planning with state space search, partial order planning, Hierarchical planning, conditional planning, Continuous and Multi Agent planning.

Learning: Introduction to Learning, Types of Learning: Learning by Induction, Rote Learning, Symbol Based Learning, Identification Trees, Explanation Based Learning, Transformational Analogy

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Formulate a problem and build intelligent agents.

CO2: Apply basic principles of AI in solutions that require problem solving, inference, knowledge representation and learning.

CO3: Analyze the problem and infer new knowledge using suitable knowledge representation schemes.

CO4: Develop planning and apply learning algorithms on real world problems.

CO5: Design by planning, learning and implementing advance techniques in Artificial Intelligence.

CO6: Create a real life and industrial problems related mini project.

TEXT AND REFERENCE BOOKS:

1. Artificial Intelligence 3e: A Modern Approach Paperback – By Stuart J Russell & Peter Norvig; Publisher – Pearson
2. Artificial Intelligence Third Edition by Kevin Knight, Elaine Rich, B. Nair – Mc Graw Hill
3. Artificial Intelligence Third Edition by Patrick Henry Winston – Addison-Wesley Publishing Company
4. M. Tim Jones, ―Artificial Intelligence: A Systems Approach (Computer Science)‖, Jones and Bartlett Publishers, Inc.; First Edition, 2008
5. A Classical Approach to Artificial Intelligence, M. C. Trivedi, Khanna Publishing House.
6. G. Luger, W. A. Stubblefield, "Artificial Intelligence", Third Edition, Addison-Wesley Longman, 1998.